THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Annual rings in a tree trunk are formed in: 1

(a) phloem **(b)** cork **(c)** secondary xylem **(d)** pith

2. The seed of wheat is a: 1

(a) monocot seed **(b)** spore **(c)** gymnosperm seed **(d)** dicot seed

3. The ovulation takes place on about: 1

(a) the 28th day **(b)** the 1st day

(c) the 7th day **(d)** the 14th day of a 28-day cycle

4. The movement of water from land to sea through rivers is: 1

(a) transpiration **(b)** condensation **(c)** run-off **(d)** evaporation

5. The area of a rectangle 50 cm by 30 cm in square metres is: 1

(a) 0.15 **(b)** 15 **(c)** 0.015 **(d)** 1.5

6. Which of the following will pass through a filter paper? 1

(a) colloidal particles **(b)** chalk powder

(c) sand **(d)** iron filings

7. Which compound has a double bond between its atoms? 1

(a) N_2 **(b)** Cl_2 **(c)** H_2 **(d)** O_2

8. The isotopes of an element differ in the number of: 1

(a) shells **(b)** neutrons **(c)** electrons **(d)** protons

9. The displacement of a body after half a circle of radius r is: 1

(a) r **(b)** zero **(c)** 2r **(d)** pi r

10. A man pushes a box of 20 kg with 50 N and the box moves at constant velocity. The friction is: 1

(a) 200 N **(b)** 20 N **(c)** 50 N **(d)** 0 N

11. The kinetic energy of a body is 50 J and its mass is 4 kg. The velocity is: 1

(a) 3.5 m/s **(b)** 12.5 m/s **(c)** 25 m/s **(d)** 5 m/s

12. A tuning fork of 512 Hz produces sound of wavelength 0.67 m. The speed is about: 1

(a) 756 m/s **(b)** 0.0013 m/s **(c)** 512.67 m/s **(d)** 343 m/s

13. Which cell structure would be most affected if a cell has no ribosomes? 1

(a) cell division **(b)** respiration only
(c) protein synthesis **(d)** photosynthesis

14. A tissue section shows living cells with unevenly thickened corners. The tissue is: 1

(a) parenchyma **(b)** sclerenchyma **(c)** xylem **(d)** collenchyma

15. An organism has a notochord, a dorsal nerve cord and gill slits at some stage. It is a: 1

(a) chordate **(b)** arthropod **(c)** mollusc **(d)** annelid

16. A student sees tiny buds on the margin of a Bryophyllum leaf. They are used for: 1

(a) respiration **(b)** vegetative reproduction
(c) storage **(d)** photosynthesis

17. Lightning contributes to the nitrogen cycle by: 1

(a) making CO2 **(b)** making ozone
(c) forming proteins **(d)** converting N2 into nitrogen oxides

18. The volume of a cuboid 2 m x 50 cm x 20 cm in cubic metres is: 1

(a) 200 **(b)** 20 **(c)** 0.2 **(d)** 2

19. To make 500 g of a 6 per cent solution, the mass of water needed is: 1

(a) 500 g **(b)** 30 g **(c)** 470 g **(d)** 494 g

20. Carbon monoxide has C : O = 3 : 4. The mass of oxygen that combines with 9 g of carbon is: 1

(a) 12 g **(b)** 6.75 g **(c)** 9 g **(d)** 16 g

21. The ion Mg^{2+} has the electronic configuration: 1

(a) 2, 6 **(b)** 2, 8, 8 **(c)** 2, 8 **(d)** 2, 8, 2

22. A car moving at 72 km/h stops in 8 s. The retardation is: 1

(a) 2.5 m/s squared **(b)** 0.9 m/s squared **(c)** 9 m/s squared **(d)** 4 m/s squared

23. A force acts on a mass of 2 kg for 4 s starting from rest and gives it a velocity of 8 m/s. The force is: 1

(a) 16 N **(b)** 8 N **(c)** 2 N **(d)** 4 N

24. A motor lifts 100 kg to 15 m in 30 s. The power is (g = 10): 1

(a) 500 W **(b)** 50 W **(c)** 1500 W **(d)** 5000 W

25. If the frequency is doubled, the pitch: 1

(a) stays same **(b)** becomes zero **(c)** becomes higher **(d)** becomes lower

26. A cell in a solution neither gains nor loses weight. The solution is: 1

(a) hypotonic **(b)** isotonic **(c)** distilled water **(d)** hypertonic

27. Cells of tissue in which the matrix is hard due to calcium and phosphorus: 1

(a) cartilage **(b)** bone **(c)** blood **(d)** areolar

28. A mammal that lays eggs is the: 1

(a) platypus **(b)** kangaroo **(c)** dolphin **(d)** bat

29. In the case of a unisexual flower, the pollination is always: 1

(a) by water only **(b)** absent
(c) self within the flower **(d)** cross-pollination (or between flowers)

30. A 5-star rated appliance is preferred because it: 1

(a) costs less always **(b)** uses less electricity
(c) is heavier **(d)** looks better

31. When a model fails in some situation, the scientist: 1

(a) defines its limits and develops a better model
(b) blames observers
(c) ignores the failure
(d) discards all science

32. To recover pure sugar from sugar solution, one uses: 1

(a) decantation **(b)** crystallisation **(c)** filtration **(d)** sublimation

33. The formula of iron(III) oxide is: 1

(a) FeO **(b)** Fe_2O **(c)** Fe_2O_3 **(d)** Fe_3O_2

34. Which of the following has the maximum valency? 1

(a) carbon (2, 4) **(b)** sodium (2, 8, 1) **(c)** oxygen (2, 6) **(d)** fluorine (2, 7)

35. In a v-t graph, a line sloping downward from 20 m/s to 0 in 4 s shows retardation of: 1

(a) 5 m/s squared **(b)** 20 m/s squared **(c)** 4 m/s squared **(d)** 80 m/s squared

36. A ball of mass 0.2 kg moving at 5 m/s is caught and stopped in 0.05 s. The average force on the hand is: 1

(a) 5 N **(b)** 20 N **(c)** 200 N **(d)** 1 N

37. A load of 150 N is raised by an effort of 50 N by a machine. The MA is: 1

(a) 1/3 **(b)** 100 **(c)** 3 **(d)** 200

38. The time between two consecutive compressions at a point is the: 1

(a) speed **(b)** time period **(c)** wavelength **(d)** amplitude

39. The smooth ER in liver cells helps in: 1

(a) cell division **(b)** detoxification
(c) photosynthesis **(d)** digestion of starch

40. Tissue between the vertebrae that acts as a cushion: 1

(a) bone **(b)** cartilage **(c)** blood **(d)** muscle

41. The largest taxonomic group is the: 1

(a) kingdom **(b)** class **(c)** phylum **(d)** species

42. The pollen grain that lands on a stigma germinates and forms a: 1

(a) new ovule **(b)** pollen tube **(c)** root hair **(d)** style

43. The albedo of a surface is 0.8. The percentage of sunlight reflected is: 1

(a) 0.8 per cent **(b)** 80 per cent **(c)** 8 per cent **(d)** 20 per cent

44. A scientist measures 12.5 g, 12.7 g and 12.6 g for the same mass. The mean is: 1

(a) 12.6 g **(b)** 12.7 g **(c)** 37.8 g **(d)** 12.5 g

45. A beam of light is visible through a mixture in a dark room. The mixture is a: 1

(a) true solution **(b)** pure substance
(c) compound **(d)** colloid or suspension

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46. Which compound has a triple bond? 1

(a) N_2 **(b)** O_2 **(c)** Cl_2 **(d)** H_2

47. $^{14}_{6}C$ is used in: 1

(a) goitre treatment **(b)** carbon dating **(c)** cancer therapy **(d)** nuclear fuel

48. The distance covered by a body after half a circle of radius r is: 1

(a) 2r **(b)** pi r **(c)** 2 pi r **(d)** zero

49. A horse pulls a cart with a force of 600 N, and the friction is 400 N. The cart's mass is 100 kg. The acceleration is: 1

(a) 10 m/s squared **(b)** 4 m/s squared **(c)** 6 m/s squared **(d)** 2 m/s squared

50. A pendulum bob of mass 0.1 kg is raised 0.2 m above its lowest point. Its speed at the lowest point is: 1

(a) 2 m/s **(b)** 0.2 m/s **(c)** 4 m/s **(d)** 10 m/s

END OF QUESTION PAPER **Total: 50 Marks**

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